一、(60%)Assume that the following sequences of instructions are executed on a five-stage pipelined datapath.

I1: ld x1,40(x6)
I2: add x2,x3,x1
I3: add x1,x6,x4
I4: sd x2,20(x4)
I5: and x1,x1,x4

1.1 Indicate dependences and their type.(RAW,WAR,WAW)

1.2 If there is no forwarding or hazard detection, insert the fewest nops to ensure correct execution.

1.3 If the processor has forwarding, but we forgot to implement the hazard detection unit, what happens when this code executes?

二、(40%)Assume you have a 1GHz processor with 2-levels of cache, DRAM main memory, and a DISK for virtual memory. The first level cache is split for instructions and data. The L1 cache is virtually addressed and physically tagged. The memory system has the following parameters:

Component	Hit Time	Miss Rate	Block Size
Level-1 Cache	1 cycle	4% for Data 1% for Instructions	64 bytes
Level-2 Cache	12 cycles(first transfer) + 1 cycle/64bits(later transfer)	2%	128 bytes
DRAM	80ns(first transfer) + 10ns/8 bytes(later transfer)	0.001%	16K bytes pages
Disk	20ms(first transfer) + 20ns/byte(later transfer)	-	-

The system includes a TLB with a miss rate of 0.1% for data. It never misses for instructions. The TLB miss penalty is 40 cycles. TLB hits take place in parallel with level-1 cache access.

2.1 What is the AMAT(average memory access time) in number of cycles for instruction accesses?

2.2 What is the AMAT in number of cycles for data accesses?